TEXORA · LECTURE SHEET

Complex Integration

Line integrals in the complex plane, Cauchy's integral theorem and formula, Taylor's and Laurent's series, singular points and residues, Cauchy's residue theorem, and real definite integrals evaluated as contour integrals.

MATH-1201 · MATHEMATICS-II · TOPIC 5 · LEVEL 1 TERM 2

1. The complex line integral

For f(z) continuous on a contour C (z = z(t), α ≤ t ≤ β),

$$\int_C f(z)\,dz = \int_\alpha^\beta f(z(t))\,z'(t)\,dt$$

It is computed by parametrising, exactly like the real line integrals of Topic 1 — but the answers behave very differently, because dz is complex: reversing the contour negates the integral, and closed contours can give non-zero answers. The crude bound $|\int f\,dz| \le M \cdot L$ (M = max |f| on C, L = length of C) is the **ML inequality**, used to kill unwanted arcs in §6.

WORKED EXAMPLES — PATH MATTERS, OR NOT

f(z) = z² from 0 to 1 + i: z^2 has antiderivative $z^3/3$ (entire), so the integral is path-independent: $[(1 + i)^3]/3 = (-2 + 2i)/3$.

f(z) = 1/z from 1 to −1: upper semicircle (z = e^{iθ}, θ: 0 → π): $\int i\,d\theta = \pi i$. Lower semicircle: $-\pi i$. Same endpoints, different answers — 1/z is not analytic on any region containing both paths' enclosed disc… the two differ by the full loop $\oint dz/z = 2\pi i$.

2. Cauchy's integral theorem

Statement: if f is analytic in a simply connected region and on its boundary contour C, then $\oint_C f(z)\,dz = 0$. (Stated and applied here, per the syllabus; proofs excluded.) Immediate consequences: integrals of analytic functions are path-independent within the region, and a contour may be deformed freely without crossing a singularity. The theorem fails exactly where analyticity fails — every non-zero closed integral in this topic is a singularity's fingerprint.

3. Cauchy's integral formula

If f is analytic inside and on C, and a is any interior point:

$$f(a) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{z-a}\,dz, \quad \text{and generally} \quad f^{(n)}(a) = \frac{n!}{2\pi i}\oint \frac{f(z)}{(z-a)^{n+1}}\,dz$$

Astonishing content: the values of an analytic function *inside* a contour are completely determined by its values *on* the contour — and every analytic function is automatically infinitely differentiable.

WORKED EXAMPLES

$\oint_{|z|=2} e^z/(z-1)\,dz = 2\pi i \cdot e^1 =$ **$2\pi i e$** ($f = e^z$, $a = 1$ inside).

$\oint_{|z|=2} z^2/(z-i)\,dz = 2\pi i \cdot (i)^2 =$ **$-2\pi i$**.

$\oint_{|z-i|=1} dz/(z-i)^2 = 2\pi i \cdot$ (derivative of 1 at i) = **0**.

4. Taylor’s and Laurent’s series

Taylor: an analytic function is a power series about any point of its region: $f(z) = \Sigma\, f^{(n)}(a)/n!\,(z-a)^n$, converging in the largest disc about a free of singularities. Standards: $e^z = \Sigma\, z^n/n!$; sin z, cos z similarly; $1/(1-z) = \Sigma\, z^n$ for $|z| < 1$.

Laurent: in an annulus between two singularities, negative powers join in: $f(z) = \Sigma_{n=-\infty}^{\infty} c_n (z-a)^n$. The coefficient c_{-1} — the coefficient of $1/(z-a)$ — is the **residue**, the single number the whole expansion is mined for.

WORKED EXAMPLE — ONE FUNCTION, TWO EXPANSIONS

$f(z) = 1/(z(z-1))$.

In $0 < |z| < 1$: $1/(z(z-1)) = -(1/z)\cdot 1/(1-z) =$ **$-1/z - 1 - z - z^2 - \ldots$**

In $1 < |z| < \infty$: $1/(z(z-1)) = (1/z^2)\cdot 1/(1-1/z) =$ **$1/z^2 + 1/z^3 + 1/z^4 + \ldots$**

Same function, different annulus, different series — always state the annulus.

5. Singular points and residues

TYPE	LAURENT SIGNATURE	EXAMPLE
Removable	No negative powers	$\sin z / z$ at 0 (value → 1)
Pole of order m	Negative powers stop at $(z-a)^{-m}$	$1/(z-2)^3$ at 2 (order 3)
Essential	Infinitely many negative powers	$e^{1/z}$ at 0

Residue recipes: at a simple pole, $\text{Res} = \lim_{z\to a} (z-a)f(z)$ — for $f = p/q$ with $q(a) = 0$, $q'(a) \neq 0$: $\text{Res} = p(a)/q'(a)$. At a pole of order m: $\text{Res} = (1/(m-1)!) \lim d^{m-1}/dz^{m-1}\,[(z-a)^m f(z)]$. Or simply read c_{-1} off a Laurent expansion.

WORKED EXAMPLES

Res of e^z/z³ at 0: e^z/z³ = (1 + z + z²/2 + z³/6 + …)/z³ → coefficient of 1/z is **1/2**.
Res of (z + 2)/(z − 1)² at the double pole z = 1: (1/1!) d/dz (z + 2) = **1**.

6. Cauchy's residue theorem and real integrals

Residue theorem: $\oint_C f(z)\,dz = 2\pi i \times$ (sum of residues at the singularities inside C). Complex integration thereby becomes bookkeeping.

WORKED EXAMPLES

Direct: $\oint_{|z|=2} (z + 2)/(z - 1)^2\,dz = 2\pi i \times \text{Res}(1) =$ **2πi** (from §5).
Unit-circle type — $\int_0^{2\pi} d\theta/(2 + \cos\theta)$: put $z = e^{i\theta}$, $\cos\theta = (z + z^{-1})/2$, $d\theta = dz/(iz)$. The integral becomes $(2/i) \oint dz/(z^2 + 4z + 1)$ around $|z| = 1$. Poles at $z = -2 \pm \sqrt{3}$; only $-2 + \sqrt{3} \approx -0.268$ lies inside, with residue $1/(2\sqrt{3})$. Result: $(2/i)(2\pi i)(1/(2\sqrt{3})) =$ **2π/√3**.
Semi-circle type — $\int_{-\infty}^{\infty} dx/(1 + x^2)$: close with the upper semicircle (its contribution → 0 by the ML estimate). Only pole inside: z = i, residue 1/(2i). Integral = 2πi × 1/(2i) = **π**. (Poles on the real axis need indentation and are excluded from this course's scope.)

7. Key points

- $\oint f\,dz = 0$ whenever f is analytic inside C (Cauchy's theorem); non-zero answers locate singularities.
- Cauchy's formula: interior values are sewn to boundary values; analytic ⇒ infinitely differentiable.
- Taylor inside discs, Laurent inside annuli; the residue is c_{-1}.
- Classify singularities by the Laurent tail: none (removable), finite (pole), infinite (essential).
- Residue theorem: replace integration by a sum over enclosed poles.
- Real integrals: trigonometric over [0, 2π] → unit circle ($z = e^{i\theta}$); rational over $(-\infty, \infty)$ → semicircle + Jordan-type estimate for the arc.

8. Practice

Answer in your own words; brief model answers follow.

1. Evaluate $\oint_{|z|=3} (\sin z)/(z - \pi/2)\,dz$.
2. Expand $f(z) = 1/(z(z + 2))$ in a Laurent series valid for $0 < |z| < 2$, and read off the residue at 0.
3. Classify the singularity at z = 0 of (a) $(e^z - 1)/z$, (b) $1/\sin z$, (c) $\cos(1/z)$.
4. Evaluate $\int_0^{2\pi} d\theta/(5 + 4\cos\theta)$ by the unit-circle method.

5. Evaluate $\int_{-\infty}^{\infty} dx/((x^2 + 1)(x^2 + 4))$ by residues.
6. State Cauchy's integral theorem precisely, and explain why $\oint_{|z|=1} dz/z = 2\pi i$ does not contradict it.

Answer key

1. By Cauchy's formula with $f(z) = \sin z$ (entire) and $a = \pi/2$ inside $|z| = 3$: $2\pi i \sin(\pi/2) = 2\pi i$.
2. $1/(z(z + 2)) = (1/(2z)) \cdot 1/(1 + z/2) = (1/(2z))(1 - z/2 + z^2/4 - z^3/8 + \ldots) = 1/(2z) - 1/4 + z/8 - z^2/16 + \ldots$, valid for $|z/2| < 1$, i.e. $0 < |z| < 2$ ($z = 0$ excluded as the singular centre). Residue at 0 = coefficient of $1/z = 1/2$.
3. (a) $(e^z - 1)/z = 1 + z/2 + z^2/6 + \ldots$ — no negative powers: **removable** (define the value 1 at 0). (b) $1/\sin z$: $\sin z$ has a simple zero at 0, so $1/\sin z$ has a **simple pole** (residue $1/\cos 0 = 1$). (c) $\cos(1/z) = 1 - 1/(2z^2) + 1/(24z^4) - \ldots$ — infinitely many negative powers: **essential**.
4. $z = e^{i\theta}$: the integral becomes $(1/i) \oint dz/((2z + 1)(z + 2))$ around $|z| = 1$. Inside poles: $z = -1/2$ only, residue $1/((2)(3/2)) = 1/3$. Integral $= (1/i)(2\pi i)(1/3) = 2\pi/3$.
5. Upper half-plane poles: $z = i$ (residue $1/((2i)(4 - 1)) = 1/(6i)$) and $z = 2i$ (residue $1/(((2i)^2 + 1)(4i)) = 1/((-3)(4i)) = -1/(12i)$). Sum of residues $= 1/(6i) - 1/(12i) = 1/(12i)$. Integral $= 2\pi i \times 1/(12i) = \pi/6$. (The arc's contribution vanishes since the integrand decays as $1/x^4$.)
6. Theorem: if f is analytic in a simply connected region, then $\oint f\, dz = 0$ round every closed contour in it. No contradiction: $1/z$ is not analytic at $z = 0$, which lies inside $|z| = 1$ — the theorem's hypothesis fails exactly there, and the $2\pi i$ is the residue (1) at that pole times $2\pi i$, as the residue theorem states.

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